CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.1 Progressive Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 20 Questions (20 Marks)

Question 1 Easy [1 mark]

A wave has a frequency of 4 GHz.

What is the period of the wave?

A 25 000 μs

B 250 ps

C 25 ns

D 2.5 ns

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The period of a wave is $T = \frac{1}{f}$
- The question states the frequency of the wave is 4 GHz which is equal to 4×10^9 Hz
  - $T = \frac{1}{4 \times 10^9} = 2.5 \times 10^{-10}$
- 2.5×10^{-10} s is the same as 250 ps

Exam Tip: You need to be confident on converting units from milli-, micro-, nano- and pico-, especially in the topic of waves. Remember that to go from a larger unit to a smaller unit you multiply and to go from larger to smaller you divide.

Question 2 Easy [1 mark]

A light with wavelength of 600 nm has an orange colour.

What is the frequency of this light?

A 500 kHz

B 500 THz

C 180 GHz

D 180 Hz

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The speed of the wave is given by the wave equation
  - $v = f\lambda$
- The speed of light is 3.0×10^{8} m s^{-1}
- The wavelength is 600 nm or 6.0×10^{-7} m
- The frequency is calculated by $f = \frac{v}{\lambda}$
  - So, $f = \frac{3.0 \times 10^{8}}{6.0 \times 10^{-7}} = 5.0 \times 10^{14}$ Hz
- There are 1×10^{12} Hz in 1 THz
  - Therefore, the frequency of this light is 5.0×10^{14} Hz = 500 THz

Question 3 Easy [1 mark]

A sound wave has a frequency of 50 Hz and a speed of 330 m s^{-1}

If two waves have a phase difference of 60˚, what is the possible distance between two points on the wave?

A 6.6 m

B 2.2 m

C 1.1 m

D 0.03 m

Mark Scheme & Explanation:

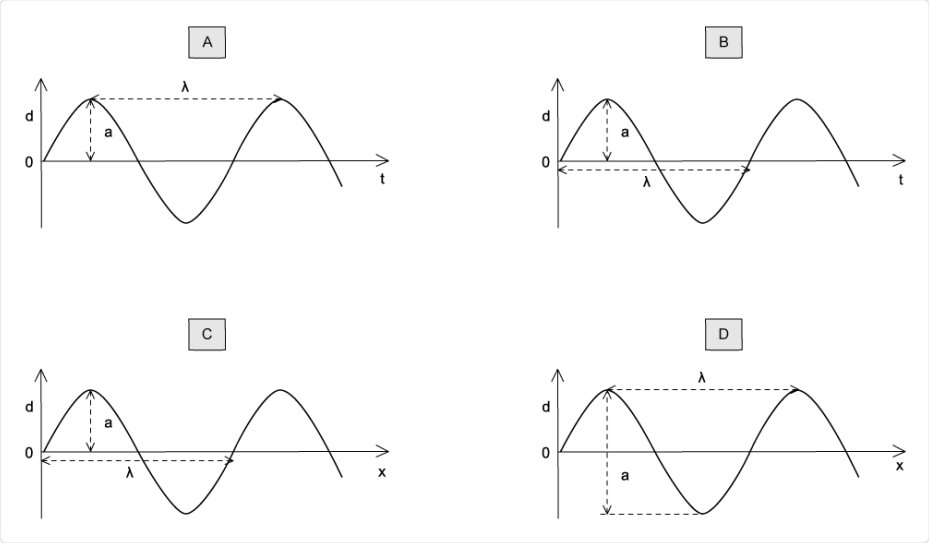
*The correct answer is **C** because:*

- The wavelength of the wave is
  - $\lambda = \frac{v}{f} = \frac{330}{50} = 6.6$ m
- 60˚ is $\frac{60^{\circ}}{360^{\circ}} = \frac{1}{6}$ of the total wavelength:
  - $60^{\circ} = 6.6 \text{ m} \times \frac{1}{6} = 1.1$ m

Question 4 Easy [1 mark]

A sound wave has a displacement *y* at a distance *x* from its source at time *t*.

Which graph shows correctly the amplitude *a* and the wavelength *λ* of the wave ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Amplitude is the maximum displacement of a particle on the medium from its resting position
- The amplitude is shown as the arrow from the resting position to the top of the peak. This is shown in answers **A, B** and **C**
- The wavelength is the minimum distance between two points on the wave oscillating in phase

A & **B** are incorrect because the x-axis on the graph is time so this would show the period of the wave not distance

D is incorrect because the amplitude shown is the displacement of one complete wave and not from the resting position

Question 5 Easy [1 mark]

The following are statements about waves.

1	Longitudinal waves can be polarised.
2	The amplitude of a wave is directly proportional to the energy transferred by the wave.
3	All electromagnetic waves travel at the same speed in a vacuum.
4	The frequency of infra-red light is greater than the frequency of ultra-violet light.

Which ones are incorrect?

A 1, 3 and 4

B 1, 2 and 4

C 1, 2 and 3

D 2, 3 and 4

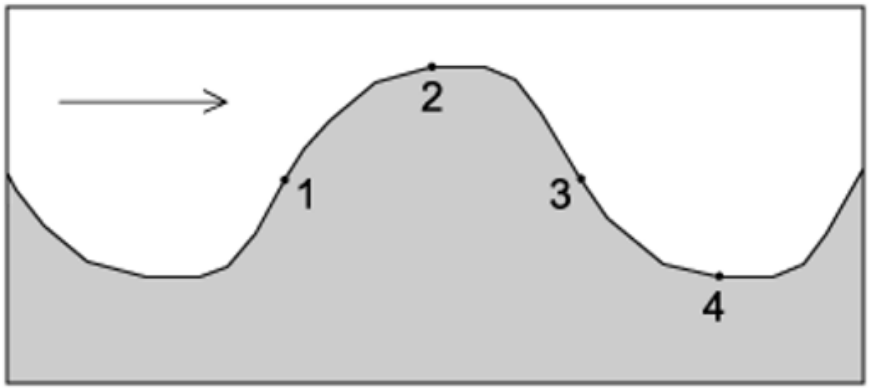
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The only statement that is correct from the options listed is that all electromagnetic waves travel at the same speed in a vacuum
- Statement 1 is incorrect as longitudinal waves can't be polarised as their particles vibrate in the same direction that the wave travels
- Statement 2 is incorrect as the square of the amplitude of a wave is directly proportional to the energy transferred. This means that doubling the amplitude results in a quadrupling of the energy
- Statement 4 is incorrect as the frequency of ultra-violet light is greater than the frequency of infra-red light
    - Range of frequencies of infrared: $3 \times 10^{12} - 4.3 \times 10^{14}$
    - Range of frequencies of ultraviolet: $7.5 \times 10^{14} - 3 \times 10^{16}$

Question 6 Easy [1 mark]

The vertical cross-section diagram shows a water wave moving from left to right.



Which points on the diagram show the water moving upwards with maximum speed?

A 1 and 3

B 3 only

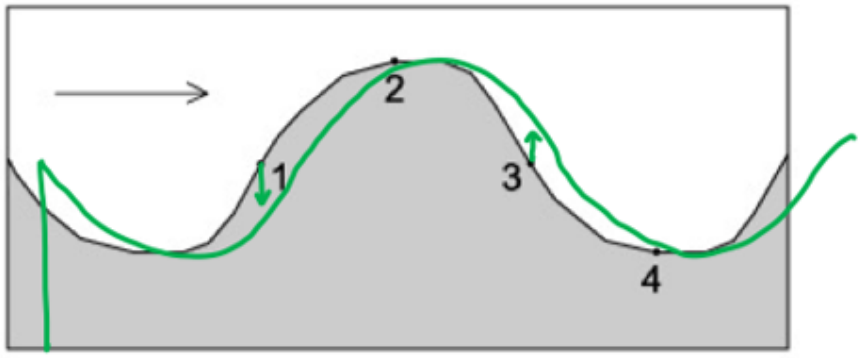
C 1 and 4

D 4 only

Mark Scheme & Explanation:

*The correct answer is **B** because:*

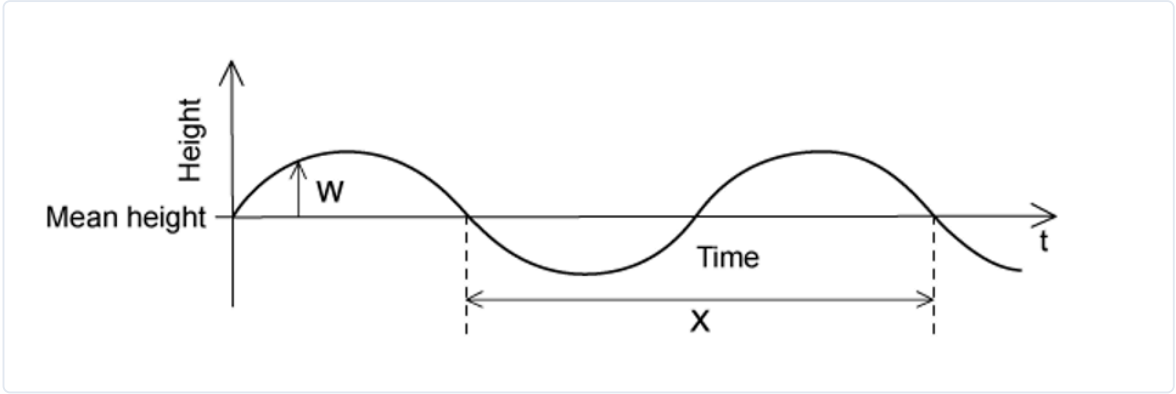
- To see where the position of the particle in the next second just shift the wave in the direction of travel
- Point 3 is the only part of the wave that is moving upwards with maximum speed



From the diagram, the particle at position 1 will be moving downwards and the particle at position 3 is moving upwards. At point 4 the particle is moving up but the question states the maximum speed so **D** is incorrect. This means the correct option is **B**

Question 7 Easy [1 mark]

The diagram below is a graph showing how the height of water at a point in a marina varies with time *t* as the waves pass the point.



What is the correct label for W and X?

	W	X
A	amplitude	period
B	displacement	wavelength
C	amplitude	wavelength
D	displacement	period

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The time is shown on the x-axis so **X** must be period as this is in units of time, not distance
- The amplitude of a wave is the maximum displacement of the wave
- **W** is not showing amplitude, **W** is displacement

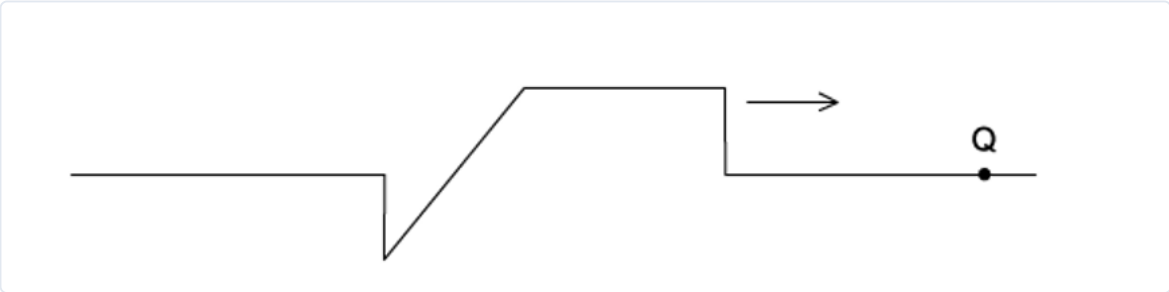
A & **C** are incorrect because W is not showing the maximum displacement, this is not the amplitude and just the displacement

B is incorrect because the units on the graph is time, if the graph was in m then this would be the wavelength

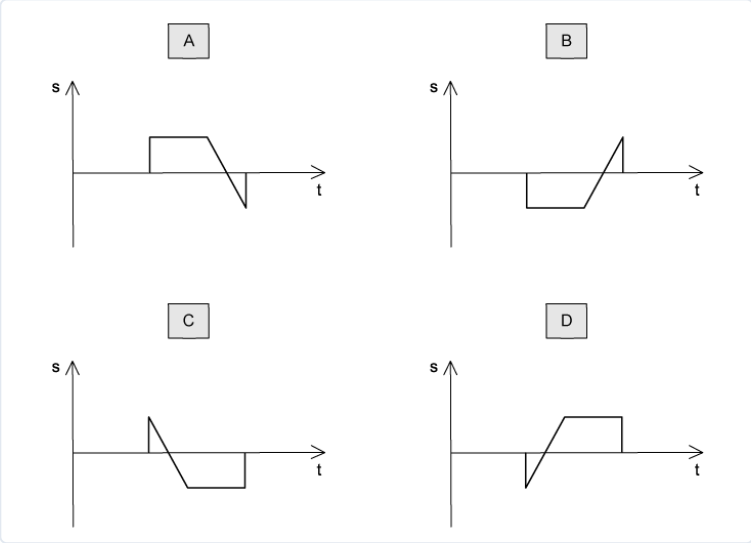
> *Always look at the units on the graph as the first instinct for students would be to assume that X is the wavelength.*

Question 8 Medium [1 mark]

The diagram shows a wave moving along a stretched rope in the direction shown.



Which of the following will be the correct graph to show the variation with time *t* of the displacement *s* of the particle Q in the rope ?



A

B

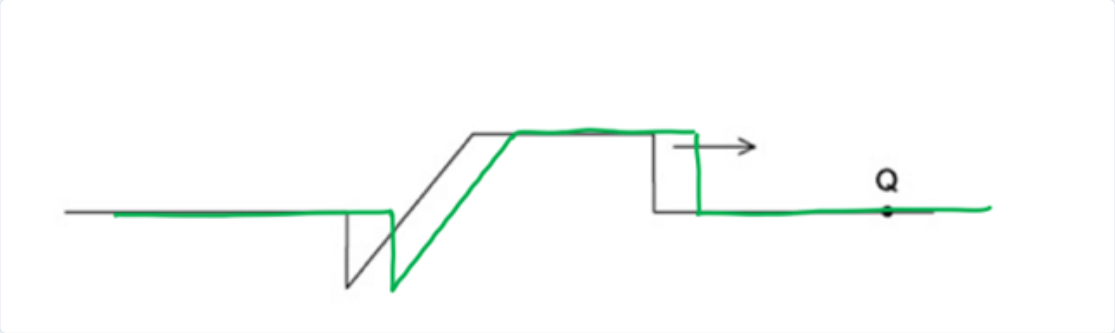
C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The wave formed along the rope is a transverse wave
- In a transverse wave, the particles oscillate perpendicular (90°) to the direction of wave travel, which is to the right in this question.
- The particles can only oscillate in a vertical motion (up and down)
- The diagram below shows the wave in an instant later, moving the wave to the right



- The particle Q will move according to the form of the wave, a sudden jump up, followed by a high value of displacement and then with a final negative peak
- The answer therefore is **A**

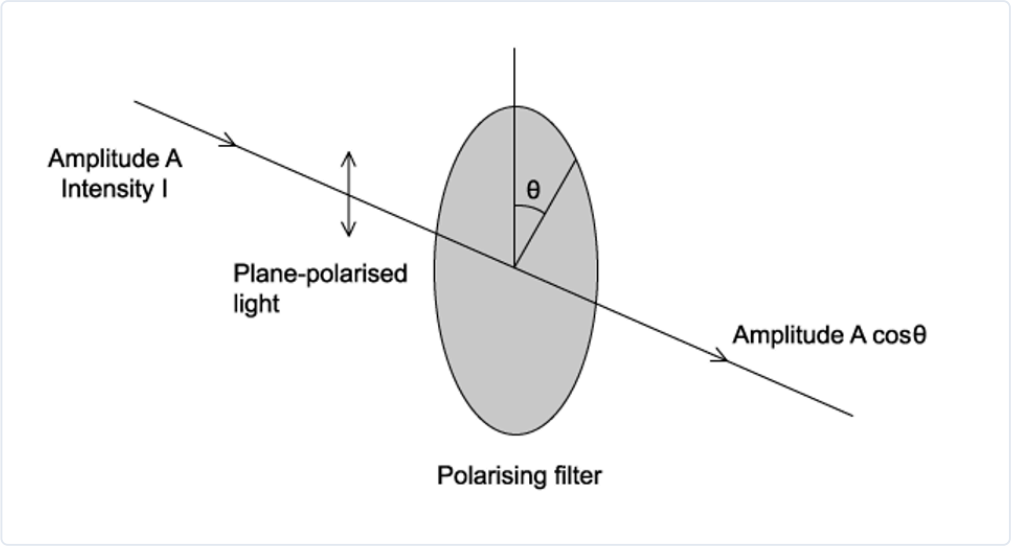
B is incorrect because the graph shows a long negative displacement with a positive peak, this is the opposite direction to what would happen to particle Q

C is incorrect because the graph shows a positive peak with a long negative displacement

D is incorrect because the graph shows a negative peak then long positive displacement which is the opposite order to what would happen to particle Q

Question 9 Medium [1 mark]

A plane polarised light of amplitude *A* is passed through a polarising filter as show in the diagram. The emerging light has an amplitude of A cos θ



If the intensity of the beam is *I*, what is the intensity of the emerging light when θ is 60.0˚?

A 0.250 *I*

B 0.500 *I*

C 0.750 *I*

D 0.866 *I*

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The intensity of light *I* is proportional to the square of the amplitude *A*
  - $A^2 \ \alpha \ I$
- The amplitude of the emerging light, let's call this $A_{em} = A \ \ cos \ \ 60°$
- Intensity of the emerging light is proportional to $A^2{}_{em} = A^2 cos^2 60° = 0.250 \ \ A^2$
  - $A^2{}_{em} \ \ \alpha \ \ I$
  - $0.250 \ \ A^2 \ \ \alpha \ \ I$

- Therefore, the intensity of the emerging light = 0.250 *I*

Question 10 Medium [1 mark]

A sound wave with an amplitude *a* of 0.2 mm has an intensity of 2.5 W m^{-2}

If a wave with the same frequency has an amplitude of 2*a* what is the intensity of the wave?

A 10.0 W m^{-2}

B 9.0 W m^{-2}

C 6.0 W m^{-2}

D 4.2 W m^{-2}

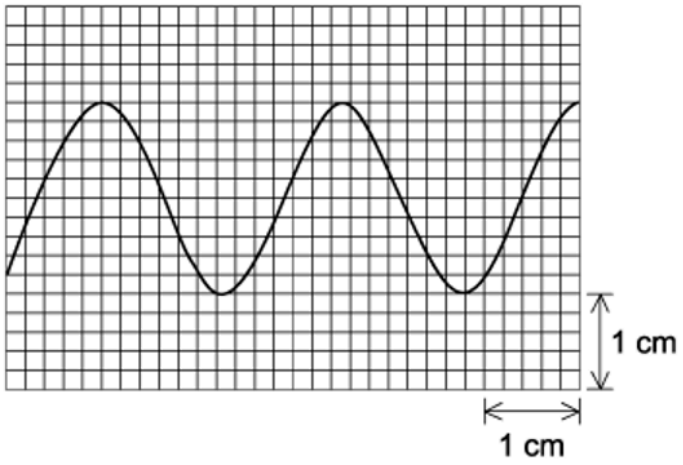
Mark Scheme & Explanation:

*The correct answer is **A** because :*

- Intensity is directly proportional to the square of the amplitude, i.e.
  - $I \ \alpha \ A^2 \quad or \quad \frac{I_1}{I_2} = \frac{{a_1}^2}{{a_2}^2}$
- Where *I* is the intensity of the wave and *a* is the amplitude of the wave
- So, we get:
  - $\frac{2.5 \ \text{W m}^{-2}}{I_2} = \frac{0.2 \ \text{mm}^2}{0.4 \ \text{mm}^2} = \frac{0.04}{0.16} = \frac{1}{4}$
  - $\therefore \ I_2 = \frac{4}{1} \times 2.5 \ \text{W m}^{-2} = 10 \ \text{W m}^{-2}$

Question 11 Medium [1 mark]

A sound wave can be displayed on a cathode ray oscilloscope (C.R.O.). The time base is set to 5 μs mm^{-1}.



What is the frequency of the sound wave?

A 5 Hz

B 100 Hz

C 5 kHz

D 100 kHz

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- From the diagram:
  - 5 divisions = 1 cm
  - So, 1 division = 0.2 cm = 2 mm
- The time base setting is 5 μs mm^{-1}
  - Therefore, 1 division = 5 μs mm^{-1} × 2 mm = 10 μs
- 1.5 waves occupy 30 divisions
  - $1.5 \quad T = 30 \times 10$
  - $T = \frac{30 \times 10}{1.5} = 200$

- The period of the wave is T = 200 μs = 2 × 10^{-4} s
- Frequency = $\frac{1}{T} = \frac{1}{2 \times 10^{-4}} = 5000 \quad \text{Hz} = 5 \quad \text{kHz}$

Question 12 Medium [1 mark]

A source of sound power P is situated in an open space. The intensity I of sound at distance r from this source is given by

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude A of the vibrating air molecules vary with the distance r from the source ?

A $A \quad \alpha \quad \frac{1}{r}$

B $A \quad \alpha \quad \frac{1}{r^2}$

C $A \quad \alpha \quad r$

D $A \quad \alpha \quad r^2$

Mark Scheme & Explanation:

The correct answer is A because:

- Intensity I is proportional to amplitude squared A^2
  - $I \quad \alpha \quad A^2$
- From the equation given in the question, we can see that
  - $I \quad \alpha \quad \frac{1}{r^2}$
- So, A^2 is inversely proportional to r^2 : $A^2 \quad \alpha \quad \frac{1}{r^2}$
  - Therefore, $A \quad \alpha \quad \frac{1}{r}$

Question 13 Medium [1 mark]

Radio waves are electromagnetic waves that travel at speed c in a vacuum with a frequency f and wavelength λ.

Which row in the table describes the wavelength and speed of radio waves with a frequency $3f$?

	Speed in a vacuum	Wavelength
A	$3c$	3λ
B	c	$\frac{\lambda}{3}$
C	c	3λ
D	$\frac{c}{3}$	$\frac{\lambda}{3}$

A

B

C

D

Mark Scheme & Explanation:

The correct answer is B

- The question states that the speed of the radio waves in a vacuum is c
- Speed is given by

$$c = f\lambda$$

- Therefore, wavelength is equal to

$$\lambda = \frac{c}{f}$$

- The speed of electromagnetic waves is constant in a vacuum so the speed will stay as c
- The frequency has increased to $3f$ so the new wavelength is

$$\lambda' = \frac{c}{3f}$$

- Comparing $\lambda = \frac{c}{f}$ with $\lambda' = \frac{c}{3f}$

$$\lambda' = \frac{1}{3} \times \frac{c}{f} = \frac{1}{3}\lambda$$

- Therefore, the new wavelength is $\lambda' = \frac{\lambda}{3}$

Question 14 Hard [1 mark]

The equation below shows the speed of *v* waves in deep water

$$v^2 = \frac{g\lambda}{2\pi}$$

Where *g* is the acceleration of free fall and λ is the wavelength of the waves. The wavelength λ and the frequency *f* of the wave were measured.

If the graph was to give a straight line through the origin which of the following graphs should be plotted?

A *f* against λ^2

B f^2 against λ

C *f* against $\frac{1}{\lambda}$

D f^2 against $\frac{1}{\lambda}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation given in the question is $v^2 = \frac{g\lambda}{2\pi}$
- The equation for the speed of the wave is $v = f\lambda$
- By substituting $v = f\lambda$ into the first equation gives
  - $f\lambda^2 = \frac{g\lambda}{2\pi}$
  - $f^2 = \frac{g\lambda}{2\pi}\frac{1}{\lambda^2} = \frac{g}{2\pi}\frac{1}{\lambda}$
- The equation of a straight line is $y = mx + c$
- A line going through the origin has a y-intercept of c = 0 so $y = mx$
  - Comparing with $f^2 = \frac{g}{2\pi}\frac{1}{\lambda} \quad with \quad y = mx$
- We can see y is equal to f^2 and x is equal to $\frac{1}{\lambda}$, so, a plot of these two quantities on a graph would give a straight line through the origin

Question 15 Hard [1 mark]

A light wave is incident normally on a surface of area *S* with an amplitude *A*. The power per unit area reaching the surface is *P*.

If the light is focused on a smaller area of $\frac{1}{3}$ S, and the amplitude of wave is doubled to 2*A* what is the power per unit area?

A 6*P*

B 12*P*

C 18*P*

D 36*P*

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The power per unit area $= \frac{power}{area}$ intensity
- When the surface area is *S* and the amplitude is *A*, the power per unit area = *P*
- The surface area has decreased to $\frac{1}{3}$ S , since intensity is inversely proportional to the area, i.e.
  - $I \ \alpha \ \frac{1}{\frac{1}{3}S} = \frac{3}{S}$
- So, when the surface area is decreased by a factor of 3 the intensity is increased by a factor of 3
- The second factor in the question is the amplitude of the wave, which is doubled to 2*A*
- Intensity is directly proportional to the square of the amplitude, i.e.
  - $I \ \alpha \ A^2$
- Doubling the amplitude increases the intensity to increase by a factor of 2^2
  - $I \ \alpha \ 2A^2 = 4A^2$
- The new power will therefore be $4 \times 3 \ \ P = 12 \ \ P$

Question 16 Hard [1 mark]

All electromagnetic waves travel at speed c in a vacuum with a frequency f and wavelength λ.

Which row in the table describes the wavelength and speed of electromagnetic waves with a frequency $\frac{f}{2}$?

	Speed in a vacuum	Wavelength
A	$2c$	2λ
B	c	2λ
C	c	$\frac{\lambda}{2}$
D	$\frac{c}{2}$	$\frac{\lambda}{2}$

A

B

C

D

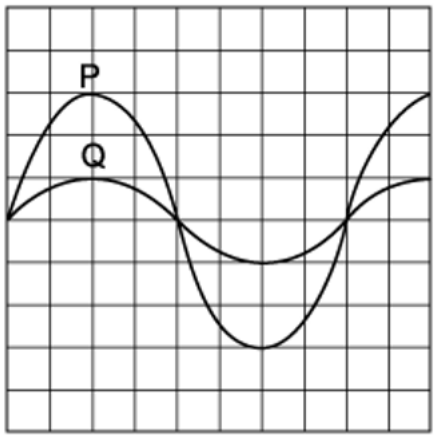
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The question states that the speed of the electromagnetic (EM) waves in a vacuum is c
  - Speed is given by $c = f\lambda$
  - Therefore, wavelength is equal to $\lambda = \frac{c}{f}$
- The speed of EM waves is constant in a vacuum so the speed will stay as c
- The frequency has decreased to $\frac{f}{2}$ so the new wavelength is $= \frac{c}{\frac{f}{2}} = \frac{2c}{f}$
- Comparing $\lambda = \frac{c}{f}$ with $\frac{2c}{f}$, we can see the new wavelength is $= 2\lambda$

Question 17 Hard [1 mark]

Sound waves can be displayed on a cathode ray oscilloscope (C.R.O.). The graph below represents two waves seen on the screen. They are labelled P and Q as shown in the diagram



What is the ratio $\frac{intensity\ \ of\ \ sound\ \ wave\ \ P}{intensity\ \ of\ \ sound\ \ wave\ \ Q}$?

A $\frac{\sqrt{3}}{1}$

B $\frac{9}{1}$

C $\frac{1}{1}$

D $\frac{3}{1}$

Mark Scheme & Explanation:

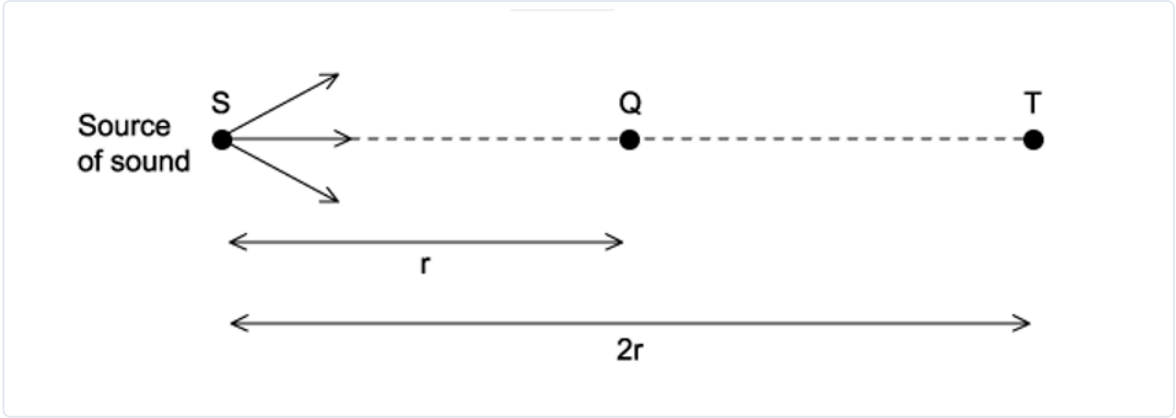
*The correct answer is **B** because:*

- Intensity *I* is proportional to amplitude squared A^2
  - $I\ \ \alpha\ \ A^2$
- The amplitude of P is 3 units and the amplitude of Q is 1 unit
  - The intensity of P = 3^2 = 9
  - The intensity of Q = 1^2 = 1
- Therefore the ratio is $\frac{intensity\ \ of\ \ sound\ \ wave\ \ P}{intensity\ \ of\ \ sound\ \ wave\ \ Q} = \frac{9}{1}$

Question 18 Hard [1 mark]

The intensity *I* of a sound at point Q is inversely proportional to the square of the distance *x* of Q from the source of the sound. This can be shown by the equation

$$I \ \alpha \ \frac{1}{x^2}$$



Molecules of air at Q are a distance *r* from S and oscillate with amplitude 8.0 μm.

Point T is a distance 2*r* from S.

What is the amplitude of oscillation of air molecules at T?

A 4.0 μm

B 2.0 μm

C 2.8 μm

D 1.4 μm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

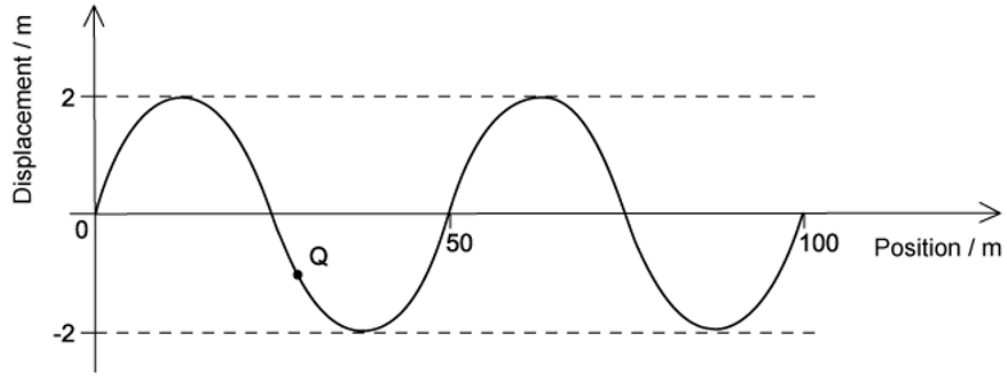
- We are told intensity is inversely proportional to the square of the distance *x* from the source so
  - $I \ \alpha \ \frac{1}{x^2}$
- When the distance is doubled this causes the intensity to decrease by a quarter
  - $I \ \alpha \ \frac{1}{2x^2} = \frac{1}{4x^2}$
- Intensity *I* is also proportional to amplitude squared A^2
  - $I \ \alpha \ A^2$

- From the question, we can see the amplitude is 8 μm, so intensity is proportional to 8^2
  - $I \ \alpha \ 8A^2$
- So, at a distance 2*r*, the intensity will be $\frac{1}{4}$and A^2 is equal to $\frac{8^2}{4}$
- Therefore, the amplitude at T is equal to $A = \sqrt{8.0^2 \div 4} = 4.0 \ \ \mu\text{m}$

Question 19 Hard [1 mark]

The graph shows a sinusoidal wave in water travelling at a speed of 8.0 m s^{-1}.

The maximum speed of the particles in the wave is $2\pi af$, where a is the amplitude and f is the frequency.



An object Q is floating on the water it has a mass of 2.5 × 10^{-3} kg.

What is the maximum kinetic energy of Q due to the wave? Assume the motion is vertical.

A 4.1 mJ

B 5.1 mJ

C 39 mJ

D 64 mJ

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The equation for the speed of a wave is $v = f\lambda$
  - So, frequency is $f = \frac{v}{\lambda}$
- From the graph, the wavelength is 50 m
- The question states that the speed is 8.0 m s^{-1}
  - $f = \frac{8}{50} = 0.16 \quad \text{Hz}$
- From the graph, the amplitude is 2 m
- The equation for the maximum speed of the particles is

- $2\pi af = 2\pi 20.16 = 0.64\pi \quad \text{m} \quad \text{s}^{-1}$

- Therefore, the maximum kinetic energy is equal to
  - $\frac{1}{2}mv^2 = \frac{1}{2}2.5 \times 10^{-3}\,0.64\pi^2 = 0.0051 = 5.1 \quad \text{mJ}$

Question 20 Hard [1 mark]

A wave has a speed of 340 m s^{-1} and a frequency of 500 Hz

If two points on a wave are 0.17 m apart, what is the phase difference?

A $\frac{\pi}{4}$ rad

B $\frac{\pi}{2}$ rad

C $\frac{3\pi}{4}$ rad

D π rad

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The equation for the speed of a wave $v = f\lambda$
- So, wavelength is $\lambda = \frac{v}{f} = \frac{340}{500} = 0.68 \quad \text{m}$
- One wavelength λ is equivalent to a phase difference of 360 degrees (which is equal to 2π rad)
- Therefore, a separation of 0.68 m (one wavelength) corresponds to a phase difference of 2π rad
- So, a separation of 0.17m corresponds to $= \frac{0.17}{0.68} \times 2\pi = \frac{\pi}{2} \quad \text{rad}$